

A GENERAL APPROACH TO MODEL HIGH-PERFORMANCE PV MODULES FOR ACCURATE ENERGY YIELD SIMULATIONS

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ABSTRACT: One of the challenges in PV module performance modeling is to obtain a good description of the I-V curves at various temperatures and illuminations, based solely on limited available data. Currently in PVsyst, this is done by calculating the parameters for the one-diode model (1DM) from datasheet information and common assumptions on low-light performance. However, this fails to reproduce high fill factors (FF), typically compromising V_{oc} accuracy. To address this, we use an evolutionary algorithm (EA) which improves parameter determination for the 1DM, extends to more sophisticated models and can use measurement variability as input. This method necessitates solely datasheet information and common low-light assumptions to reproduce IEC 61853-1 measurements. We demonstrate its benefits on both measured PERC modules and synthetic high-FF scenarios. On measured data from a low-FF module, the EA lowers the mean efficiency error in reproducing IEC 61853-1 data compared to the current method. To test the EA on high-FF devices, we generate synthetic data from a reference recombination model informed by solar-cell literature. In this case too, the EA reduces power error and greatly improves V_{oc} reproduction, which is further enhanced when switching from 1DM to the reference model itself.

Keywords: PV Module Modeling, PVsyst

1 INTRODUCTION

Accurate modelling of PV module I-V curves is essential for reliable energy yield simulations. The one-diode model (1DM), as implemented in PVsyst, remains an industry standard due to its simplicity and accuracy to represent PERC PV modules. PVsyst uses an analytical method to determine the model parameters based solely on the information available in datasheet, completed by observed technological trends for low-light performance.

However, for modern high-performance devices with elevated fill factors (FF), this approach often compromises open-circuit voltage accuracy, limiting its predictive power. Moreover, the origin of those high FF within crystalline silicon (c-Si) technology is well understood through more sophisticated models incorporating several recombination mechanisms impacting in parallel the I-V behaviour. These developments call for parameter evaluation methods that go beyond fixed assumptions and can flexibly adapt to new device characteristics.

To overcome the limitations arising at high FFs, we explore two approaches: 1) Change the determination method for the model parameters evaluation from datasheet information. 2) Investigate the potential benefit of a different I-V-parametrization that is established in literature and known to be able to describe high FF.

In particular, we propose an evolutionary algorithm (EA) framework that can adapt to both the conventional 1DM and recombination-based models. This method necessitates solely datasheet data and common low-light assumptions to reproduce IEC 61853-1 measurements. We demonstrate its benefits on both measured PERC modules and synthetic high-FF scenarios.

2 METHOD

We first summarize the current PVsyst I-V parameterization and a reference recombination-based model from the solar cell literature state-of-the-art. This model is commonly accepted to describe the origin of high FF in record efficiency device [1] and we will therefore use it to generate synthetic data in the next sections.

Then, we describe the current calculation procedure of PVsyst and the EA optimization approach.

2.1 Current PVsyst I-V Parametrization

We describe here the parametrization of the 1DM as it is implemented in the current version of PVsyst (8.0). As in any 1DM parametrization, the external current I is a balance between the photogenerated current and the current losses in the first diode and shunt resistor,

$$I = I_{ph} - I_0 \left(e^{\frac{q(V+IR_s)}{\gamma k_B T}} - 1 \right) - \frac{V + IR_s}{R_{sh}}$$

where the term $V_{int} = V + IR_s$ is the internal voltage of the solar cells prior to the voltage drop across the series resistance R_s . The equivalent circuit of this equation can be seen in Figure 1 (a).

The photogenerated current I_{ph} is assumed to depend linearly on the temperature and irradiance

$$I_{ph}(G, T) = \frac{G}{G_{ref}} I_{ph,ref} [1 + \mu_{Isc}(T - T_{c,ref})]$$

where $I_{ph,ref}$ is the photogenerated current under the standard test conditions (STC) temperature $T_{c,ref} = 25^\circ\text{C}$ and irradiance $G_{ref} = 1000 \text{ W/m}^2$. μ_{Isc} is the short-circuit current temperature coefficient. Similarly, the saturation current of the diode I_0 depends on the temperature as

$$I_0(T) = I_{0,ref} \left(\frac{T}{T_{ref}} \right)^3 \exp\left(\frac{qE_g}{\gamma k_B} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right)\right),$$

where for crystalline silicon (c-Si) the bandgap E_g is set to a fixed value of 1.12 eV.

Next, based on experimental observations [2], PVsyst assumes an exponential behaviour of the shunt resistance with irradiance

$$R_{sh}(G) = R_{sh,Base} + [R_{sh}(0) - R_{sh,Base}] \times \exp\left(-R_{sh,exp} \left(\frac{G}{G_{ref}} \right)\right)$$

with

$$R_{sh,Base} = \frac{R_{sh}(STC) - R_{sh}(0) \exp(-R_{sh,exp})}{1 - \exp(-R_{sh,exp})}$$

debatable, Homer says this is STC value.

Based on the measurement campaign led in [2], the two additional degrees of freedom $R_{sh,exp}$ and $R_{sh}(0)/R_{sh}(STC)$ ratio takes remarkably fixed values for a given PV technology, leaving only the shunt resistance at 1000 W/m², $R_{sh}(STC)$, as unknown parameter.

Finally, PVsyst assumes a linear dependence on temperature of the diode ideality factor γ

$$\gamma(T) = \gamma_{ref} + \mu_\gamma(T - T_{ref}),$$

where μ_γ is an additional unknown parameter.

In summary, PVsyst 1DM has 9 parameters:

$$I_{ph,ref}, \mu_{I_{sc}}, I_0, \gamma, \mu_\gamma, R_s, R_{sh}(STC), R_{sh,exp}, \text{ and } R_{sh}(0).$$

We additionally assume $R_{sh,exp} = 5.5$ and $R_{sh}(0) = 4 R_{sh}(STC)$ for c-Si, which reduces the number of unknown parameters to 7.

2.2 Reference recombination model for high-FF

We explicitate here the reference parametrization of the I-V curve informed by solar cell literature on electron/hole recombination mechanisms. As for the 1DM, the model is still described as a current balance equation

$$I = I_{ph} - I_{aug} - I_{rad} - I_{SRH} - I_{sh},$$

where the different terms are defined below and the equivalent circuit diagram is shown in Figure 2 (a).

For the description of the intrinsic recombination, we follow the latest parametrization proposed in [3] [4]. In this formalism the Auger recombination current

$$I_{aug} = qW(g_{ehh}C_{p0}(p^2n - p_0^2n_0) + g_{eeh}C_{n0}(pn^2 - p_0n^2))$$

and the radiative recombination current

$$I_{rad} = qW(1 - f_{PR})B_{low}B_{rel}(np - n_0p_0)$$

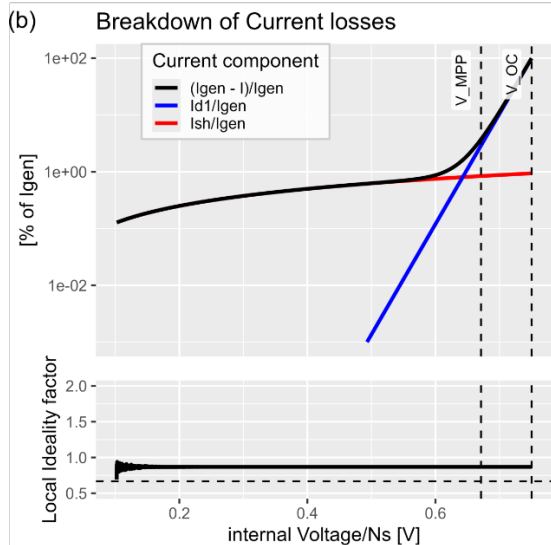
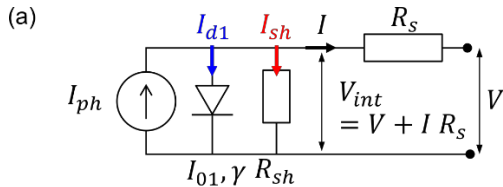


Figure 1: (a) Equivalent circuit diagram of the one diode model. (b) Breakdown of current losses for typical model parameters on logarithmic scale and local ideality factor derived from the slope of the top black curve.

depend explicitly on the electron and hole carrier concentration n and p , respectively, and the c-Si wafer thickness W . Those quantities are related to the equilibrium concentration n_0 and p_0 by

$$n = n_0 + \Delta n \text{ and } p = p_0 + \Delta n$$

where Δn is the excess charge concentration created by the excitation (voltage and illumination), related to the intrinsic carrier concentration n_i by

$$n_0p_0 = n_i^2.$$

Finally, in the case of n-type wafer, we have the approximation $n_0 = N$ (and $p_0 = N$ for p-type), where N is the base doping concentration of the wafer. We follow [4] for the value of the coefficients C_{p0}, C_{n0}, B_{low} and the remaining terms are defined in the same publication.

Note that the carrier concentrations are directly related to the internal voltage by

$$n_i^2 \exp\left(\frac{qV_{int}}{k_B T}\right) = pn \approx \left(\frac{n_i^2}{N} + \Delta n\right)(N + \Delta n).$$

Therefore, an increase in Δn results directly in an increase of V_{int} .

Within the description of [3] [4], n_i depends both on temperature and carrier concentrations. A commonly accepted parametrization is [5] [6]

$$n_i(T, n, p) = 9.653 \times 10^9 \times \left(\frac{T[K]}{300}\right)^{1.706} \exp\left(-\frac{E_G(T, n, p)}{2k_B T}\right)$$

where the function $E_G(T, n, p)$ represents the effect of the bandgap narrowing effect at increased carrier concentration. Note the resemblance with the 1DM's I_0 temperature dependance.

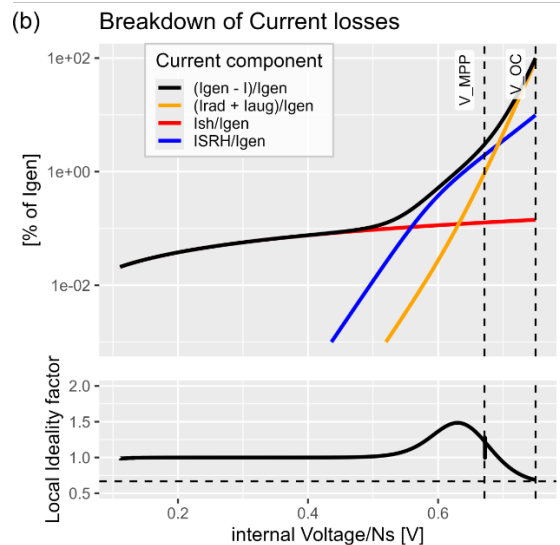
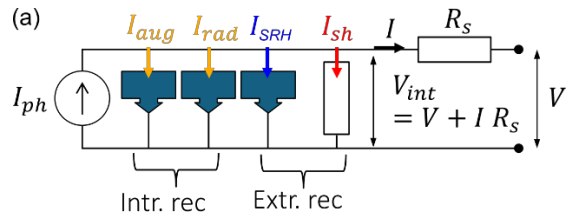


Figure 2: (a) Equivalent circuit diagram of the recombination model. (b) Current losses breakdown for typical model parameters and local ideality factor derived from the slope of the top black curve.

For the description of the extrinsic recombination, we replace the single diode term with the Shockley-Read-Hall equation which uses the same formalism as presented in this section and, in the assumption of single defect level close to mid-gap [1], can be approximated as

$$I_{SRH} = q W \frac{pn - n_i^2}{\tau_{SRH}(n + p)}$$

where τ_{SRH} is the effective lifetime of the excess charge created by the excitation (voltage and illumination) and W the thickness of c-Si wafer.

The shunt current is still given by

$$I_{sh} = \frac{V + IR_s}{R_{sh}}$$

and we keep the same assumptions for the irradiance and temperature dependance of R_{sh} and I_{ph} . This also sums up to 9 unknown parameters

$I_{ph,ref}, \mu_{Isc}, \tau_{SRH}, W, N, R_s, R_{sh}(STC), R_{sh,exp},$ and $R_{sh}(0)$

where the same assumptions can be made on $R_{sh,exp}$ and $R_{sh}(0)$, which reduces the number of unknown parameters to 7 again.

The motivation to use this model is now described. It is a well-known fact that the STC efficiency of an ideal c-Si solar cell (i.e. with no extrinsic recombination nor shunt or series resistance) is limited by radiative and Auger recombination to a value of 29.4% while the FF is limited to 89.26%. Note that those numbers remained unchanged up to the second digit in the last decades [7] [8] [4]. The FF limit is explained by the strong increase of the Auger recombination at higher carrier concentrations (p^2n and pn^2 terms), and therefore at higher voltages, increasing gradually the intrinsic recombination current from the maximal power point (MPP) until the open-circuit conditions (OC). This is shown in Figure 2 (b). Another way to understand this, is that if one were to approximate by an effective single diode model representing the intrinsic recombination mechanisms, it would be required to introduce a voltage dependent diode ideality factor which decreases towards a value of 2/3 as the voltage increases, making the I-V curve's shape more squared and increase the FF. This justifies Auger & radiative parameters being not the SRH...

2.3 Current PVsyst Parameter Calculation

In the current PVsyst implementation (8.0), the direct calculation method of the 1DM parameters requires at least 9 independent input data to be well defined. The 6 first are (almost always) specified in the PV module datasheet.

First, I-V curve is constrained to pass by three points defined by the STC values: (V_{mpp}, I_{mpp}) , $(V_{oc}, 0)$, and $(0, I_{sc})$. Note that it is not specified that the first point is the maximum power point, and therefore the P_{mpp} value obtained by the model can be slightly different than the one of the input data.

Second, μ_{Isc} is used directly as input parameter and the power temperature coefficient μ_{Pmpp} needs to be reproduced (mainly by adjusting the parameter μ_γ). Note that the temperature coefficient of the open circuit voltage μ_{Voc} is not used in this calculation (its usage is restricted to compute in a separate way the V_{oc} at low temperature for system sizing and norm safety purpose).

Third, as mentioned above, two additional constraints can be set by fixing $R_{sh,exp} = 5.5$ and $R_{sh}(0) = 4 R_{sh}(STC)$, validated for c-Si technology.

The final constraint is the relative efficiency loss under

R_s fits greatly impacts low-ir behavior?

low-light conditions (200 W/m², 25 °C) compared to STC

$$\text{Rel. eff.} = 1 - \eta_{low} / \eta_{STC}$$

This strongly helps to determine R_s since the power loss goes as $P_{loss} \sim I^2 R_s$. Unfortunately, this information is almost never present in product datasheets and PVsyst assumes by default a value of -3% when no information is available based on previous experimental campaigns [2]. Since then, with the constant reduction in series resistance in PV manufacturing, driven by improvements in contact layer resistance, metallic grid conductivity and increased number of busbars, this relative efficiency has been decreasing consistently on average to values below -5% [9]. A lower bound of about -6.8% for this value can be estimated based on the measured series resistance of some best state-of-the-art c-Si solar cell [1]. Therefore, in the present study, recognizing the relative low light efficiency value of -3% as optimistic with currently available PV modules, we use a value of -4.5% when no other information can be assumed.

The first set of constraints defines 1DM parameters with a degree of freedom, leaving the series resistance free up to a maximum value $R_{s,max}$. The additional constraint imposed by the relative low-light efficiency fixes this value. However, in the case of high FF, not all values of relative low-light efficiency can be achieved while respecting the other constraints [10]. In these cases, PVsyst will artificially increase the V_{oc} value until it is possible to respect all the constraints, such as the MPP which is the first relevant information to preserve for accurate energy yield simulations. Such an example is presented in Figure 3, where we choose the extreme example of the above-mentioned certified record solar cell [1], leading to an inaccurate reproduction of the V_{oc} . In comparison, fitting those data using either the reference recombination model of section 2.2 or a two-diode model allows an accurate reproduction of the V_{oc} . This will be discussed further in the next sections.

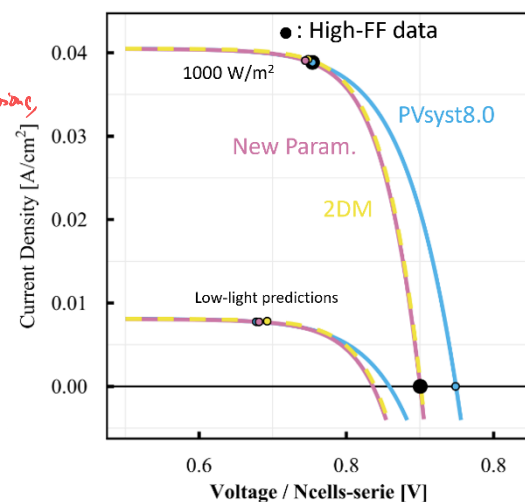


Figure 3: High FF solar cell certified data from [1] and corresponding I-V curve estimation from PVsyst 8.0 1DM, a two-diode model and the intrinsic recombination model ("New Param").

2.4 Alternative Parameter Evaluation

To circumvent the limitations of the current 1DM parameters calculation of PVsyst, we propose a fitting method based on an Evolutionary Algorithm (EA). EA are popular to handle non-linear optimization problems and

easily adaptable to various problem formulations. They are notably used by some PV module characterization centers in order to produce .PAN files reproducing as closely as possible IEC 61853-1 measured data. In this work, we use a **penalty-based differential evolution algorithm** which was shown to beat other evolutionary algorithms in reproducing the original two-diode model parameters of synthetically generated data [11].

Our EA fit works by optimizing the set of model parameters of the 1DM. Note that it can also be adapted to find the parameters of the recombination model of section 2.2. Based on the typical available information in product datasheet and low-light performance assumption to reproduce, we write a fitting objective function that needs to be minimized by an optimal set of parameters. This function is written as

$$F_{obj}(Params) = \frac{1}{\sigma_{P_{mpp}}} (P_{mpp} - P_{mpp,fit}) / P_{mpp} + \frac{1}{\sigma_{V_{mpp}}} (V_{mpp} - V_{mpp,fit}) / V_{mpp} + \frac{1}{\sigma_{V_{oc}}} (V_{oc} - V_{oc,fit}) / V_{oc} + \frac{1}{\sigma_{I_{sc}}} (I_{sc} - I_{sc,fit}) / I_{sc} + \frac{1}{\sigma_{\mu_{P_{mpp}}}} (\mu_{P_{mpp}} - \mu_{P_{mpp,fit}}) / \mu_{P_{mpp}} + \frac{1}{\sigma_{\mu_{I_{sc}}}} (\mu_{I_{sc}} - \mu_{I_{sc,fit}}) / \mu_{I_{sc}} + \frac{1}{\sigma_{\mu_{V_{oc}}}} (\mu_{V_{oc}} - \mu_{V_{oc,fit}}) / \mu_{V_{oc}} + \frac{1}{\sigma_{Rel. eff}} (Rel. eff - Rel. eff_{fit}) / Rel. eff$$

where the subscript “fit” indicate the respective values computed by the chosen I-V model for a given set of parameters, the elements without subscripts are the measured values and the σ 's represent their respective standard deviation.

The latter is of importance because some input data have much stronger experimental variations. For example, in the case of data obtained from IEC 61853-1 reports, three similar modules are measured. By analyzing reports from several test centers, we observed that while the STC values variation is well below 1% among the three modules, the variation in $\mu_{I_{sc}}$ is of the order of 10%. Likewise, when the low-light efficiency is unknown, we can more generally assume a value of $-4.5\% \pm 2\%$, covering widely all the observable values reported in the literature [9]. **Another advantage of this method is that it allows for the use of the $\mu_{V_{oc}}$ information.** Note that any other available information that can be computed from an I-V model could be used in this objective function and it could therefore be adapted to reproduce all the measured elements of an IEC 61853-1 report.

3 TEST SCENARIOS

In this section, two case studies are used to compare the parameter evaluation method of the current PVsyst model and the EA fit. In the first case, we focus on **reproducing IEC 61853-1 certified data from a real PERC PV module, with a conventional FF value of 78.4%, based solely on the above-cited available information.**

In the second case, we repeat the operation with data

synthetically generated from the reference recombination model of section 2.2. This emulates a PV module with very high FF (85.7%) and very low relative low-light efficiency (-6.8%), which represents closely the best performance we could expect one day from c-Si PV module and serves as an extreme case to test the presented methods.

3.1 Evolutionary Algorithm Fit for One Diode Model and Standard Fill Factor

Figure 4 shows how the 1DM EA fit reproduces the measured data of the PERC device at all temperatures and irradiances available in its IEC 61853-1 report. The efficiency and the V_{oc} are well fitted at 1000 W/m², but a small discrepancy at low-light level can be seen. This is explained by the fact that the real measured relative low light efficiency for this device ($-6.2\% \pm 1.2\%$) was assumed unavailable and the default value of $-4.5\% \pm 2\%$ was used instead according to our hypothesis on commonly available data.

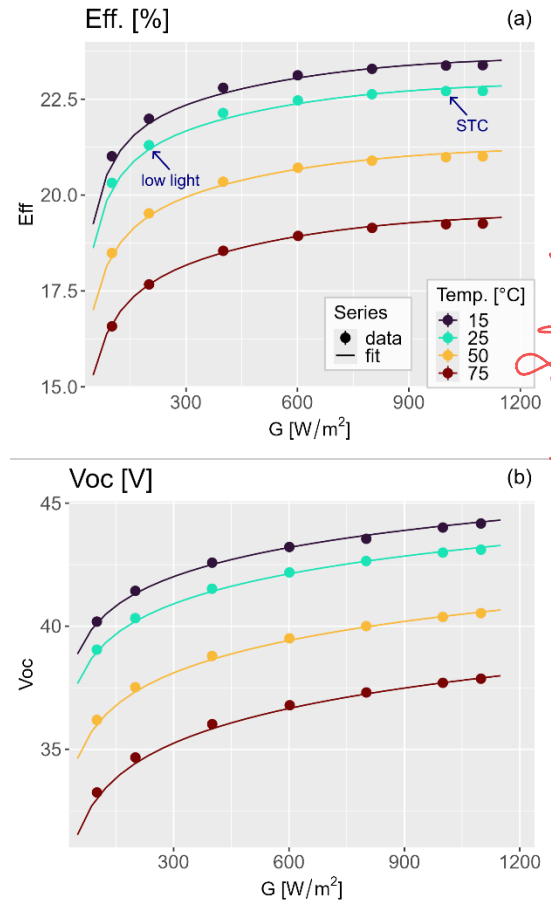


Figure 4: Reproduction of measured IEC 61853-1 data from the EA fit for a conventional FF (78.4%, PERC) based only on limited input (P_{mpp} , V_{mpp} , V_{oc} , I_{sc} , $\mu_{I_{sc}}$, $\mu_{V_{oc}}$, $\mu_{P_{mpp}}$) and conventional assumption on low light efficiency ($-4.5\% \pm 2\%$) [9].

Figure 5 shows how the measured parameters are reproduced by (a) the current PVsyst method and (b) the EA fit with the 1DM. Both methods are in good agreement and reproduce the IEC MPP data within 2.5%. The EA fit reduces the error further, close to the tolerated standard deviation of each PV module characteristic. It provides better reproduction of the MPP and V_{oc} data, their

temperature coefficients, and reduces error across all temperature and irradiance conditions. This is due to the advantages of the EA fit, which leverages μ_{Voc} information, variations in μ_{Isc} , and (often unknown) low-light efficiency, whereas PVsyst discards the first and assumes a fixed value for the others.

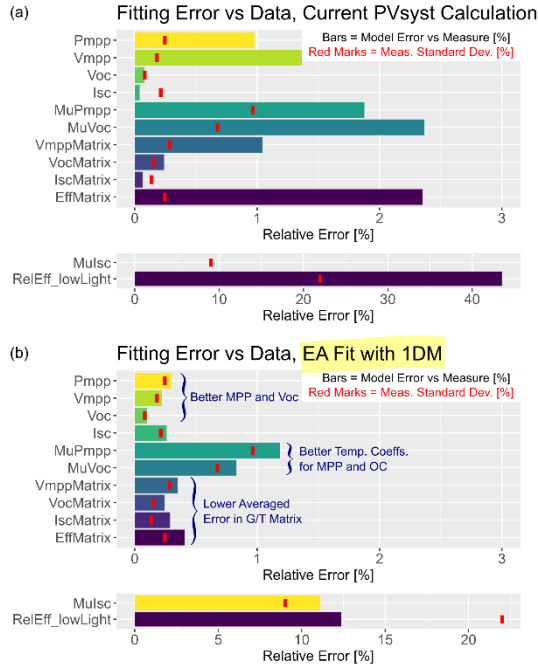


Figure 5: Error from (a) the PVsyst calculated model and (b) the EA fit model in reproducing the IEC 61853-1 data. Pmpp, Vmpp, Voc and Isc refer to the STC conditions, MuPmpp, MuVoc and MuIsc to the temperature coefficients calculated at 1000 W/m² from the data at the different temperatures, VmppMatrix, VocMatrix, IscMatrix and EffMatrix to the corresponding average error in reproducing the data over all the temperatures and illumination, and RelEff_lowLight is the ratio of the efficiency at 1000 W/m² and 200 W/m² (at 25 °C).

3.2 Fitting Synthetic Data for Hypothetical High Fill Factor

Synthetic data for STC values, temperature coefficients and relative low-light efficiency were generated using the recombination model of section 2.2, using input parameters reproducing closely the certified measured data of [1]. Random errors on the temperature and irradiance were added to emulate measurement errors. This scenario raises a caveat, since we are attempting to fit data with a relative low-light efficiency of -6.8% while assuming 4.5% for a modern PV module.

Figure 6 shows the relative error in reproducing P_{mpp} and V_{oc} at temperatures and irradiances relevant to PV system simulations for three approaches. The errors in reproducing P_{mpp} are small for each approach and mainly driven by the low light assumptions. The EA Fit improves even further the results as it does not impose a strict value for the low light efficiency. As expected for this test scenario, the error on V_{oc} by the PVsyst calculation is large, up to 4%. Very interestingly, the EA fit manages to

reduce this error below 0.5%, allowing a very good reproduction of both P_{mpp} and V_{oc} while keeping the same 1DM parametrization as PVsyst. Using the same model parametrization as the reference one used for generating data yields even lower error on the V_{oc} . However, the error on the P_{mpp} becomes larger. We explained this by the fact that this parametrization is less flexible and cannot allow a relative low light efficiency too far from the requested -4.5% without creating discrepancies on the other parameters. Therefore, in the case of unknown low light efficiency, the 1DM parametrization seems the most adapted to reproduce well high FF PV module behavior over all conditions of interest.

4 SUMMARY AND OUTLOOK

Accurately modelling PV module I-V curves across operating conditions remains a key challenge for yield simulations, especially for high-performance devices with high fill factors. The current PVsyst approach, based on direct parameter calculation for the one-diode model, provides reliable estimates but compromises open-circuit voltage reproduction when faced with high-FF technologies.

In this work, we introduced an evolutionary algorithm to improve parameter evaluation for the one-diode model. On measured data from a standard PERC module, the EA fit reduced errors in maximum power point, open-circuit voltage, and temperature coefficients compared to the current method, while robustly handling measurement variability and unavailable low-light performance. For a synthetic high-FF case, the EA successfully lowered V_{oc} reproduction errors from several percent to below 0.5%, demonstrating its ability to capture the voltage-dependent recombination mechanisms characteristic of state-of-the-art c-Si devices.

The results show that evolutionary algorithm provides a flexible framework that can estimate accurately PV performance at all relevant temperature and irradiance conditions based solely on available datasheet information and common low-light assumptions. It does so with greater accuracy than the current PVsyst deterministic calculation especially when low-light efficiency is unknown (leveraging μ_{Voc} and typical low-light trends). Importantly, even though the EA method already increases significantly the accuracy of the 1DM evaluation, it enables a straightforward extension to more advanced parametrizations.

This general framework lays the groundwork for more accurate PVsyst .PAN files, ensuring robust energy yield simulations for the next generation of PV technologies. Future work will focus on validating the approach with high-FF commercial modules. The possible exploitation of the full IEC matrix also offers promising prospects. The sensitivity to missing or inconsistent data (e.g. product datasheet v.s. certified IEC report) should be studied in more details. Finally, this case study focused exclusively on PV modules of a single power class. Other power classes could not be evaluated due to the lack of certified measurements; however, it would be valuable to investigate them once manufacturers provide standardized data.

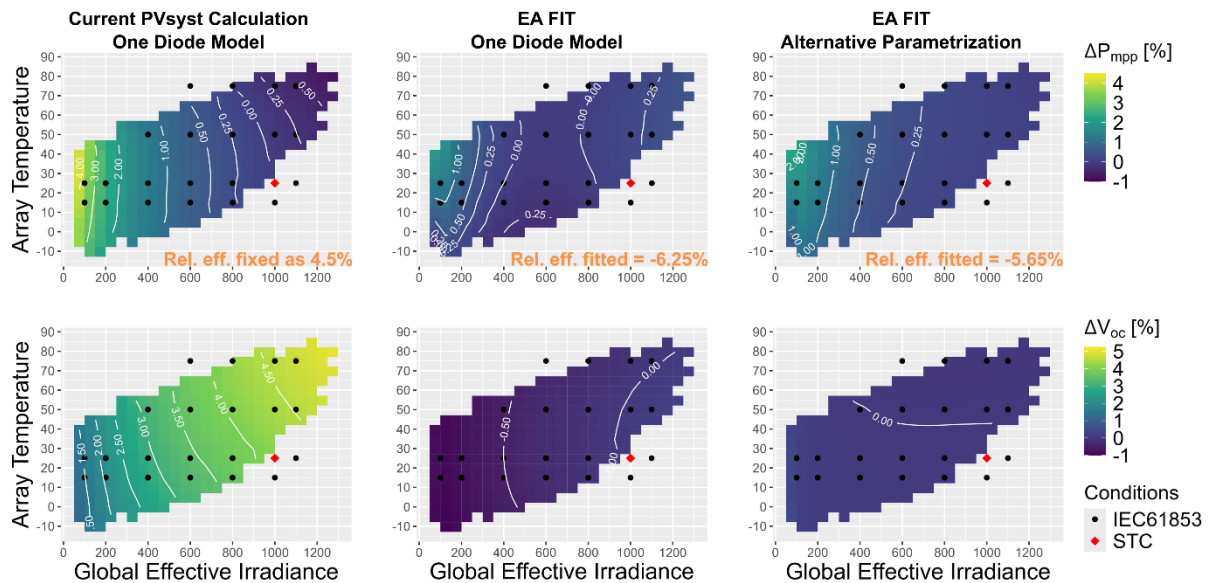


Figure 6: Relative error on Voc and Pmpp in reproducing the synthetic data for relevant temperatures and irradiances, either using the current PVsyst calculation with the 1DM, the EA fit with the 1DM, or the EA fit with the reference model parametrization (same as the one used for generating the synthetic data, denoted as “alternative parametrization”). The black dots references represent the mandatory measurement conditions within the IEC 61853-1 norm and the STC are marked by a red square.

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